Comparison Between Bladder and Rectal Dose using Single and Three Channel Vaginal Applicator in HDR Electronic Brachytherapy

A Thesis report submitted to Department of Medical Physics and Biomedical Engineering (MPBME), Faculty of Postgraduate Studies, Gono Bishwabidyalay in partial fulfillment of requirement for the degree of Masters of Science (M.Sc.) in Medical Physics and Biomedical Engineering.

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It is hereby declared that this project report or any part of it has not been submitted elsewhere for the award of any degree or diploma.

Signature of the candidate



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LIST OF ABBREVIATIONS

BT	Brachytherapy
EB	Electronic Brachytherapy
2D	2-Dimensional
3D	3-Dimensional
LDR	Low Dose Rate
MDR	Medium Dose Rate
PDR	Pulse Dose Rate
BEV	Beam's Eye View
CT	Computed Tomography
DVH	Dose-Volume Histogram
EBRT	External Beam Radiation Therapy
OAR	Organ-At-Risk
MLC	Multileaf Collimators
PDD	Percentage Depth Dose
RT	Radiation Therapy
TPS	Treatment Planning System
AP	Antero-Posterior Field
PA	Postero-Anterior
IORT	Intraoperative Radiation Therapy
ICBT	Intracavity Brachytherapy
CTV	Clinical Target Volume
PTV	Planning Target Volume
DVO	Dose Volume Optimizer



ABSTRACT

Abstract

The analysis focuses on dose distributions and adherence to prescribed constraints for the vagina, bladder, and rectum. The findings indicate significant deviations from intended dose levels across all assessed organs. In the single-channel approach, the vagina experienced substantial dose escalation, surpassing 4000 cGy, while the bladder and rectum received doses well above the prescribed limits. Similarly, the three-channel applicator showed patterns of dose escalation in both bladder and rectum, exceeding intended constraints by significant margins. These escalated doses highlight the need for meticulous treatment planning and optimization to ensure safer procedures for these critical organs. The comparison underscores consistent trends of escalated doses in the bladder and rectum, emphasizing the necessity for comprehensive dose optimization in HDR electronic brachytherapy.

The comparison between single-channel and three-channel vaginal applicators in HDR electronic brachytherapy unravels critical insights into organ-specific dose analyses. The vaginal, bladder, and rectal doses, examined within the context of prescribed constraints, offer crucial perspectives on the efficacy and adherence of this treatment modality. For the single-channel approach, the vaginal analysis demonstrated discrepancies in doses, notably exceeding intended levels. While meeting some constraints, the maximum dose far surpassed the intended dose, posing potential risks. The bladder and rectal analyses revealed doses significantly exceeding prescribed limits, emphasizing the importance of meticulous planning for safer treatments.

With the three-channel applicator, similar trends of escalated doses in the bladder and rectum were observed, indicating substantial deviations from intended constraints. Both organs exhibited maximum doses well beyond prescribed levels, highlighting the necessity for dose optimization to mitigate potential complications. The comparison between single and three-channel vaginal applicators consistently showed escalated doses in the bladder and rectum, underscoring the heightened risk of adverse effects. These findings stress the urgency for comprehensive treatment planning and optimization to align delivered doses more closely with prescribed constraints, ultimately enhancing the safety and efficacy of HDR electronic brachytherapy for patients in these anatomical regions.

The comparison between single and three-channel vaginal applicators in HDR electronic brachytherapy reveals substantial deviations from prescribed constraints for the vagina, bladder, and rectum. Both approaches showed escalated doses in critical organs, emphasizing the need for meticulous treatment planning and dose optimization to ensure safer and more effective brachytherapy procedures.



Chapter 1: Introduction

1.1 General Introduction

The success of radiation therapy in treating cervical cancer relies on delivering a high radiation dose directly to the tumor while sparing the surrounding healthy tissues to some extent. The curative potential of radiation therapy, particularly in managing cervical cancer, is significantly augmented by the use of intracavitary brachytherapy. This form of treatment is often employed either as a standalone approach or, more commonly, in conjunction with external beam radiation therapy (EBRT), surgery, and/or chemotherapy. In a typical radiotherapy department, approximately 10-20% of all radiotherapy patients undergo brachytherapy. Usually combined with EBRT, brachytherapy aims to elevate the radiation dosage to the area at highest risk for tumor recurrence, such as the original tumor site or the tumor bed post-surgical resection [1].

The American Brachytherapy Society (ABS) strongly advocates that cervical cancer radiation treatment, whether with or without chemotherapy, should encompass brachytherapy as an integral component [2].

Cancer treatments vary, with radiotherapy being among the most prevalent methods, alongside surgery and chemotherapy. Radiotherapy involves local treatment using ionizing radiation to eliminate cancer cells. This radiation can be administered externally, irradiating the tumor from multiple angles outside the body, or internally, through brachytherapy where radioactive sources are placed inside the body. Internal therapy, particularly brachytherapy, reduces radiation exposure to healthy tissues, thus minimizing the patient's overall deterioration [3].

High-dose-rate brachytherapy (HDR brachytherapy) constitutes a form of internal radiotherapy wherein the tumor is briefly exposed to high-dose-rate radiation. Applicators are inserted into or near target tissues within the patient's body. A radioactive source, often iridium-192, travels through these applicators via an afterloader device with remote source control. This source remains in predefined locations within the applicators for a specified dwell time, delivering a precisely defined radiation dose to the tumor while minimizing exposure to surrounding organs. Post-treatment, the applicators are removed [4].

Other common brachytherapy techniques include low-dose-rate (LDR) and pulse dose-rate (PDR) brachytherapy, differing primarily in their dose rates, measured in Gray per hour (Gy/h), based on the strength of the radioactive source. HDR, exclusively performed with afterloading devices and boasting a high dose rate (>12 Gy/h), completes treatments in minutes per session, avoiding damage to healthy tissues. In contrast, LDR involves a lower dose rate (0.4 to 2 Gy/h), where the source is implanted for several days or permanently. Finally, PDR administers the dose over multiple days in pulses, with a rate between 0.5 to 2 Gy/h [3, 4].

This study exclusively focuses on HDR brachytherapy due to its prevalence. Used as a boost treatment after external beam radiation therapy or as a standalone therapy, HDR brachytherapy finds application in treating uterine, breast, bronchus, lung, esophagus, and prostate cancers. Compared to LDR, HDR brachytherapy offers numerous advantages. Apart from its dosimetric superiority owing to the stepping source, it presents practical and economic benefits: reduced hospital staff exposure to radioactivity, a single radiation source, no patient isolation between sessions, and shorter treatment times, alleviating patient burden [5]. Intracavitary brachytherapy remains the most widely used technique in cervical cancer patients [5].

Optimizing doses in brachytherapy is not a new concept but has gained widespread attention due to technological advancements, particularly in computing power and imaging modalities like CT and MRI. While various applicator systems and prescription methods have been used for cervix cancer, variations of the Manchester system have been predominant. However, limitations of these methods have been well-documented. The advent of CT and MRI-compatible applicators, alongside 3D treatment planning, enables detailed visualization of target volumes and critical structures, improving dose accuracy. Studies have highlighted underestimations of bladder and rectum doses predicted by the Manchester system [5, 6]. With imaging advancements, manual optimization methods have evolved into inverse planning simulated annealing (IPSA), employing mathematical algorithms to iteratively achieve an optimal solution by minimizing the objective function [7].

Recently, the American Brachytherapy Society (ABS) published recommendations for adjuvant vaginal HDR brachytherapy post-hysterectomy, outlining dose prescription, optimization guidelines, and dose fractionation schedules. While there's consensus to treat the upper third of the vagina as a target due to reported treatment failures in this region, the optimal applicator choice varies widely. The traditional single-channel vaginal cylinder, due to its radial dose distribution, offers limited optimization possibilities based on patient anatomy. Multichannel vaginal cylinder (MCVC) brachytherapy aims to enhance vaginal brachytherapy capabilities, bridging the gap between single-channel cylinders and interstitial brachytherapy (ISBT) applicators. MCVC's additional channels support more conformal dosimetry, potentially reducing bladder and rectum doses compared to single-channel cylinders. The study aims to retrospectively compare the dosimetric advantages of a Miami MCVC versus a single-channel cylinder in intracavitary vaginal HDR brachytherapy post-hysterectomy, assessing target and organs-at-risk (OARs) volumes, constraint definitions, coverage, dose homogeneity, and OAR sparing [5, 7].

1.2 Brachytherapy for the Treatment of Cervical Cancer

Cervical cancer is the fourth most frequent cancer worldwide and the most prevalent among women in Africa, Asia, and South America, with an estimated annual incidence of 528,000 cases and 266,000 annual cervical cancer deaths [10]. For early-stage disease (typically FIGO stages IB-IVA), the current standard of care involves a combination of external beam radiotherapy (EBRT) and intracavitary brachytherapy (ICBT). EBRT is initially administered to address regional and pelvic disease while reducing the tumor volume. Subsequently, ICBT is used to intensify the dose delivered to the affected area while minimizing the dose to surrounding healthy tissue. High-dose rate (HDR) brachytherapy (i.e., > 12 Gy/hr) is a common form of ICBT for cervical cancer treatment, with a typical fractionation scheme of 5.5-7 Gy per fraction for 5 fractions. This combined approach has led to successful cervical cancer treatment for a significant patient population [11].

Brachytherapy is particularly well-suited for cervical cancer treatment due to several reasons. Radiation emitted from brachytherapy sources is isotropic, diminishing with the square of the distance. This characteristic enables therapeutic delivery to diseased tissue while minimizing exposure to surrounding normal tissues. Moreover, precise localization of the vagina and uterus poses concerns in EBRT, whereas placing the source in vivo aids accurate dose delivery to the intended tissues. However, a notable downside of ICBT is the inevitable exposure of the bladder and rectum to radiation. Consequently, delivering therapeutic radiation doses via ICBT is often constrained by these two organs, leading to potential complications such as proctitis, fistulas, cystitis, and bladder and rectal ulceration [12].

1.3 HDR in Cervical Cancer

Cancer of the uterine cervix is among the malignancies effectively treated with radiotherapy. High dose-rate (HDR) brachytherapy stands as a vital modality for treating gynecological (GYN) cancers. Notably, it allows for highly conformal dosing to the target aided by image guidance. For cervical cancer patients, this approach has translated into improved local control and enhanced cancer cure rates in comparison to alternative techniques [13]. Combining beam treatment with brachytherapy augments tumor dosing while minimizing exposure to organs at risk. HDR brachytherapy has gained acceptance as a primary cervical cancer treatment. Its advantage over Low Dose Rate (LDR) treatment lies in its capacity to treat a larger number of patients, especially those with larger tumors in regions with limited resources, as seen in countries like Bangladesh.

The utilization of Iridium 192 (Ir-192) sources for HDR brachytherapy is due to the ease of manufacturing smaller-sized brachytherapy applications [14]. Remote motorized insertion of

sources allows medical personnel to monitor patient treatment from an adjacent room, ensuring safety during the technique. Hollow applicators, needles, or catheters inserted into the tumor connect to the radiation material via transfer tubes, safely administered within the HDR afterloader. Remote-controlled brachytherapy can employ low dose rate (LDR), medium dose rate (MDR), or high dose rate (HDR) techniques. While the International Commission on Radiation Units and Measurements (ICRU) defines HDR as Gy/hr, typical doses administered by brachytherapy units range from 100 to 300 Gy per hour.

The efficiency of HDR is evident in its quick treatments lasting only a few minutes, feasible on an outpatient basis with minimal risk of applicator movement and patient discomfort. Modern HDR afterloaders utilizing single-stepping sources allow for dose distribution optimization by adjusting dwell times at each position. However, it's crucial to note that while optimization enhances dose distribution, it cannot compensate for poorly placed implants. HDR typically comprises multiple fractionated treatments, though it can be administered as a single intraoperative treatment if doses to normal tissues are managed through displacement or shielding [15].

1.4 Applicator Evolution

Brachytherapy for cervical cancer can be performed using an intracavitary, interstitial, or a combination approach. Intracavitary brachytherapy involves placing the radioactive source via an applicator through the vaginal cavity into the uterine cavity. On the other hand, interstitial brachytherapy utilizes catheters (small tubes) inserted into residual disease areas through a transparent vaginal approach. The choice of technique depends primarily on disease extent and anatomy. Determining the appropriate approach is crucial, starting at the time of diagnosis. Intracavitary brachytherapy remains the most commonly practiced form, with a wide variety of commercially available applicators existing [16].

The precision of source positioning to sub-millimeter accuracy and control over dwell time at these positions have led to innovations in applicator design and treatment planning. Early applicator designs consisted of a single catheter and a rigid structure, causing discomfort for patients. Moreover, the rectum curves at a distance of approximately 10cm from the anal verge, limiting treatments below this curve [17]. While effective for treating circumferential tumors, the single channel applicator falls short in shielding the rectal tissue from high doses near the cancer site. Efforts are made to minimize dose exposure to the bladder wall, prostate, and bone marrow, prioritizing the reduction of radiation to these organs at risk [17, 18].

Subsequently, applicators of diverse shapes, sizes, and varying numbers of channels were developed. This evolution significantly improved tissue sparing of organs at risk (OARs).

Figure 1 demonstrates how the Miami applicator struggled to conformally treat the tumor without exposing the bladder to a high dose. This scenario resulted in a fraction of the organ receiving unintended high doses, particularly 50% of the prescription dose, highlighting the need to minimize bladder exposure to radiation [17, 18]. Comparatively, newer applicators showed substantial enhancement in sparing OARs. In a dose comparison study, the multichannel applicator demonstrated a 50% reduction in bone marrow exposure, 70% reduction in healthy rectal tissue exposure, and complete avoidance (100%) of radiation to the bladder and prostate in terms of 50% dose exposure to these OARs [17].

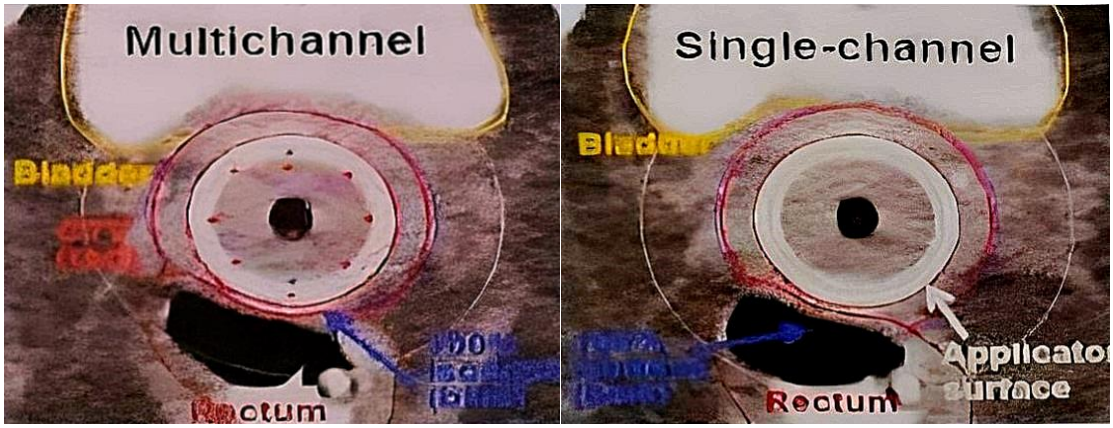


Figure 1: Example of dose distribution using the single channelled applicator in left side, while the right dose distribution is using a multiple channelled applicator for cervical carcinoma [18].

1.5 Objectives of the Study

1.5.1 General Objective

The General purpose of this Study is to retrospectively compare, via 3D inverse Planning optimization. the potential dosimetry advantages of a multichannel vaginal applicator versus a single channel one, in intracavitary vaginal HDR brachytherapy after hysterectomy. This comparison is made by analysing dose distributions to target and organs at risk (OARs) volumes.

1.5.2 Specific Objectives

- To compare diametrical optimization between Singh channel and Miami Multi channel applicator using dose volume histograms (DVH) treated carcinoma of cervix with intra-cavitary brachytherapy.
- To compare the bladder and rectal doses between two applicators in cervical cancer.



- To evaluate the dosimetric superiority using various dose indices parameters in HDR brachytherapy planning for cervical carcinoma.
- To evaluate use of Miami multi-channel applicator and single-channel applicator (SC) to improve vaginal disease coverage and dose minimization in bladder and rectal point.

1.6 Scope of the Study

The academic work has been done in Department of Medical Physics & Biomedical Engineering, Gono University, Savar, Dhaka.

This study has been done in the Gazi Cancer Care Center, Dhaka, Bangladesh for the electronic brachytherapy. Data are taken from Xoft Axxent Electronic Brachytherapy system for medical use.

1.7 Literature Reviews

Over the last decade, significant advancements have occurred in the field of imaging, treatment techniques, construction of treatment machines, and radiation measuring instruments. One exciting technique involves using a sealed radioactive source placed inside or adjacent to the area requiring treatment [1]. Research for this literature review was conducted via Medline (via PubMed), Scholar, and manual searches. Manual searches involved identifying references in articles and specific searches focused on treatment plan optimization and main study concepts. The literature review comprises a comprehensive analysis of several studies:

In the study by Lucas C. Mendez et al. (2017), a comparison was made between dosimetry performance of multi-channel vaginal (MCVC) and industrial brachytherapy (ISBT) for vaginal tumors. The research analyzed the dosimetric outcomes for patients treated with ISET and targeted intermediate (IRCTV) or high-risk (HRCTV) clinical target volumes, showcasing differences in dose distribution between ISBT and MCVC plans [16].

Umesh et al. (2017) focused on comparing 3D dose volume histogram (DVH) parameters of bladder and other organs at risk during high-dose-rate intracavitary brachytherapy (HDR-ICBT) for cervical cancer. The dosimetry study evaluated different bladder filling protocols, highlighting the impact of bladder volume on DVH parameters [19].

Tomas P. et al. (2017) investigated inverse treatment planning methods for cervical therapy. Their study involved comparing manually optimized planning with various inverse planning methods, analyzing dosimetric parameters for cancer patients treated with magnetic resonance imaging (MRI)-guided high dose rate (HDR) brachytherapy [20].

Yuhong Zhou et al. (2017) conducted a study on self-developed vaginal cylinder applicators and Auto Brachy, an automated HDR brachytherapy planning suite. They aimed to improve treatment planning efficiency and reduce plan quality variations. The research demonstrated the efficacy of automated treatment planning with clinically acceptable quality [21].

Matthew M. Harkenrider et al. (2017) published a report on risk stratification and treatment recommendations for vaginal brachytherapy in the postoperative management of endometrial cancer patients. Their literature review highlighted the efficacy and recommendations outlined by relevant societies to guide practitioners in VBT usage [22].

Kehwar et al. (2016) investigated the applicability of dosimetry systems and guidelines in high dose rate brachytherapy of cervical carcinoma. The study discussed the Manchester system and ICRU recommendations, revealing differences in coverage between these systems and ABS HR-CTV based dose prescriptions [23].

Ram Abhinav Kannan et al. (2015) performed a study comparing manual optimization with inverse planning simulated annealing (IPSA) in intracavitary brachytherapy for cervix carcinoma. Their findings showed significant improvement in target volume coverage with IPSA planning compared to manual optimization [24].

Samuel M. Shin et al. (2015) evaluated the use of multi-channel applicators to improve vaginal disease coverage achievable by single-channel applicators and compared it to Syed plan simulation. The study demonstrated increased tumor coverage and reduced doses to organs at risk with multi-channel applicators [25].

Antonio L. Damato et al. (2015) conducted a study to investigate the dose distributions associated with a novel balloon sleeve placed over a standard cylinder applicator. Their findings highlighted optimal dose coverage with the novel sleeve configurations [26]

Yasir A. Bahadur et al. (2014) investigated the dosimetric advantage of multichannel vaginal applicators over single-channel ones in HDR brachytherapy for endometrial carcinoma. Their research showcased reduced doses to organs at risk with multichannel applicators [27]

Ileana Fatima et al. (2013) carried out a study to develop a standardized procedure from simulation to treatment delivery for the Miami applicator, aiming to increase planning consistency and reduce errors. Their approach resulted in increased planning consistency and met dosimetric criteria [28].

William Shaw et al. (2013) studied establishing a generalized equivalent uniform dose (gEUD) -based prescription method for Image Guided Brachytherapy (IGBT) for cervix carcinoma patients. The research compared different planning approaches and demonstrated equivalence within normal tissue constraints [29].



These studies collectively provide insights into various techniques, dosimetry methods, and planning approaches in brachytherapy for gynecological cancers.



Chapter-2 Theoretical Background

2.1 Brief Scenario of Gynaecological Cancer in Bangladesh

Cancer is anticipated to emerge as an increasingly significant cause of both morbidity and mortality in Bangladesh in the forthcoming decades (Author, Year). The estimated incidence, presently standing at 12.7 million new cases, is projected to surge to 21.4 million by 2030 (Author, Year). Alarmingly, cancer claims more lives than tuberculosis, HIV, and malaria combined. During the 1960s, roughly a quarter of the world's diagnosed cancer cases were reported in low and middle-income countries. By 2010, this burden had escalated to nearly 55%. At the current growth rate, by 2050, the likelihood of contracting cancer in developing countries will increase to 50-60% in a person's lifetime, marking a rise from 650,000 to 2.2 million new cases annually [33].

Bangladesh, encompassing 147,570 km² (56,977 square miles), stands as one of the most densely populated nations and faces this challenge starkly. GLOBOCAN 2008 reports a five-year cancer survival rate of 291.2 thousand [34]. Annually, there are 141.1 thousand new cases and 103.3 thousand cancer-related deaths, totaling 124.8 thousand cases, including both new and existing ones. The most prevalent cancers, regardless of gender, comprise lung, breast, cervix uteri, lip and oral cavity, and esophagus [35]. Bangladesh's statistical bureau ranks cancer as the sixth leading cause of death. The International Agency for Research on Cancer indicates a rise in cancer-related fatalities in Bangladesh from 7.5% in 2005 to 13% in 2030. Notably, lung and oral cancers predominate in males, while breast and cervical cancers prevail among females.

However, Bangladesh grapples with a severe shortage of radiation therapy machines, hospital beds, trained oncologists, medical radiation physicists, and technologists. As the population burgeons and life expectancy climbs, the nation foresees a surge in cancer patients. Addressing this surge necessitates a substantial government commitment to both preventive measures and curative services [36].

Globally, cervical cancer stands as the fourth most prevalent cancer among women and takes the lead in Africa, Asia, and South America, with an estimated annual incidence of 528,000 cases and 266,000 deaths [35]. Despite effective screening methods, limited screening programs make cervical cancer a persistent issue in developing countries, including Bangladesh, where it ranks as the second most common malignancy among women in terms of occurrence and mortality. Although prior studies have explored cervical cancer incidence and treatment in Bangladesh, there remains a scarcity of published data on treatment outcomes. Erecting a robust cancer care infrastructure in Bangladesh presents formidable challenges. Data collection in resource-constrained countries has proven onerous, prompting suggestions for innovative technological applications. Moreover, the lack of physical infrastructure and insufficient resources, including an inadequate number of radiation therapy units, exacerbate the challenge [36].

2.2 Current Radiation Treatment Modalities for Cervical Cancer

The most common radiation therapy modalities in clinical use today can be classified *Ito categories, namely external beam radiation therapy and brachytherapy.

2.2.1 External Beam Radiation Therapy

External beam radiation therapy (EBRT) constitutes a treatment approach where a beam is generated outside the patient's body, contrasting the use of internal materials typical in brachytherapy. The electromagnetic type can involve photons, whereas particulate types, such as electrons or protons, are often utilized in clinical settings. The choice of radiation energy and type hinges on factors like the target's depth and clinical availability, among others. Photon energies typically range between 1 to 25 MV in clinical practice and are often administered using a linear accelerator [12].



Figure 2: Linear Accelerator for External Bean Radiation Therapy

2.2.2 Brachytherapy

Brachytherapy involves placing radioactive material within or very near the target area. Various classifications of brachytherapy exist, including dose rate, treatment site, or method of administration—interstitial versus intracavitary, for instance. Among the radioactive isotopes utilized for brachytherapy, Iridium-192 stands prominent. This modality proves ideally suited for treating cervical cancer due to its capacity to deliver therapeutic doses to diseased tissue while

minimizing exposure to surrounding normal tissues. However, a drawback of intracavitary brachytherapy (ICBT) remains the unavoidable dose exposure to the bladder and rectum, often limiting the therapeutic radiation delivery due to potential complications like proctitis, fistulas, cystitis, and bladder and rectal ulceration [12].

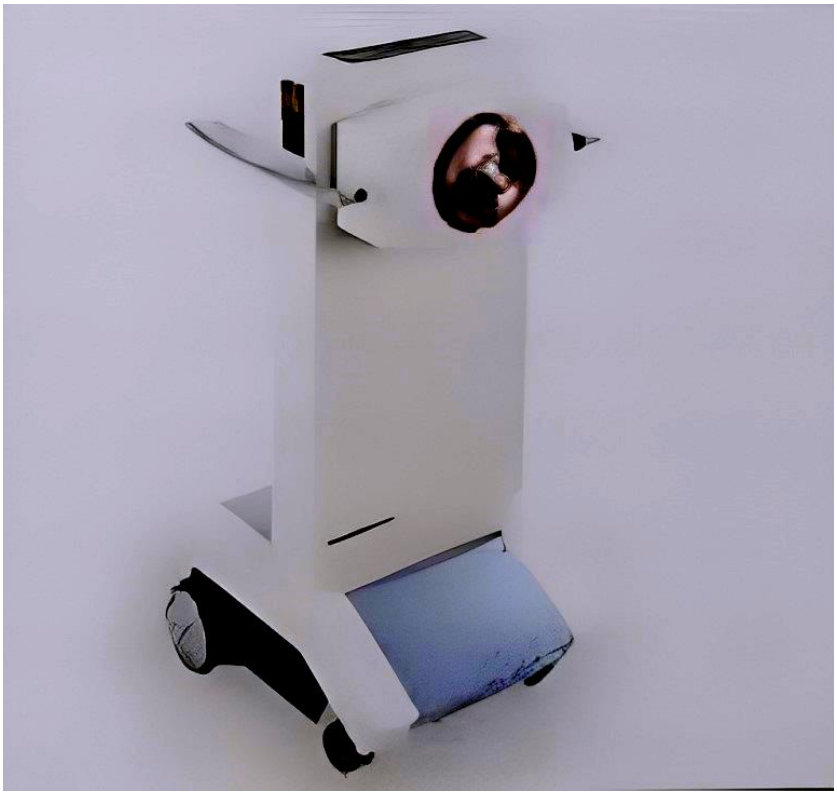


Figure 3: High Dose Rate Brachytherapy Machine

2.3 Introduction of Conformal, 3D Planning in Brachytherapy

Brachytherapy holds paramount importance in comprehensive radiotherapy for cervical cancer. The evolution of appropriate device concepts in modern radiation therapy has been pivotal in advancing Brachytherapy [37]. The advent of 3D planning in external beam radiotherapy (EBRT) signifies a monumental stride toward personalized patient treatment, improved clinical outcomes, and reduced toxicity [38].

In 3D brachytherapy treatment planning, accurate localization of the target and at-risk areas, along with applicator reconstruction, relies on modern imaging modalities. Previously, the reliance on plain X-ray films for gynecological and intracavitary applications limited the efficacy of 3D reconstruction for target localization and dose evaluation. Specifically, for intracavitary brachytherapy in primary cervical cancer, discrete anatomical points have traditionally been utilized for documenting distribution to the anatomy. However, this method, termed 'projectional reconstruction' (PRM), suffers limitations in defining anatomical volumes like PTV and OARs due to its dependence on X-ray films [39]. Consequently, the conventional use of X-ray

radiographs lacks the necessary correlation between anatomy and dosimetry, prompting a shift towards 3D sectional imaging modalities like CT, MR, and ultrasound (US) for treatment planning purposes. This transition is increasingly prevalent, especially in Western countries, replacing conventional methods [39].

The integration of imaging-based planning, primarily seen in prostate cancer treatments, where US imaging aids pre- and intraoperative planning, alongside CT for post-planning, has catalyzed developments in imaging-based planning applicable to various brachytherapy applications. The utilization of 3D imaging enables anatomy-adapted implantation, treatment planning, and optimization in brachytherapy. Moreover, CT, MR, and USG serve as guidance tools during needle insertion, ensuring a high level of safety and real-time verification of needle position concerning anatomy [40].

In the context of cervical cancer, 3D brachytherapy planning not only offers established dosimetric advantages over 2D planning but also clinical benefits. These include confirming applicator placement accuracy, reducing critical OAR doses in cases of a small cervix, accounting for sigmoid colon dose, and enhancing coverage for extensive disease while preserving organs at risk [40]. Notably, 3D planning has contributed significantly to the diagnosis and prevention of unsuspected perforations during tandem placement, thereby avoiding overtreatment of uterine tissue [40]. The enhancement in diagnostic capabilities due to 3D planning has led to improved patient care, highlighting its pivotal role in contemporary brachytherapy practices [38].

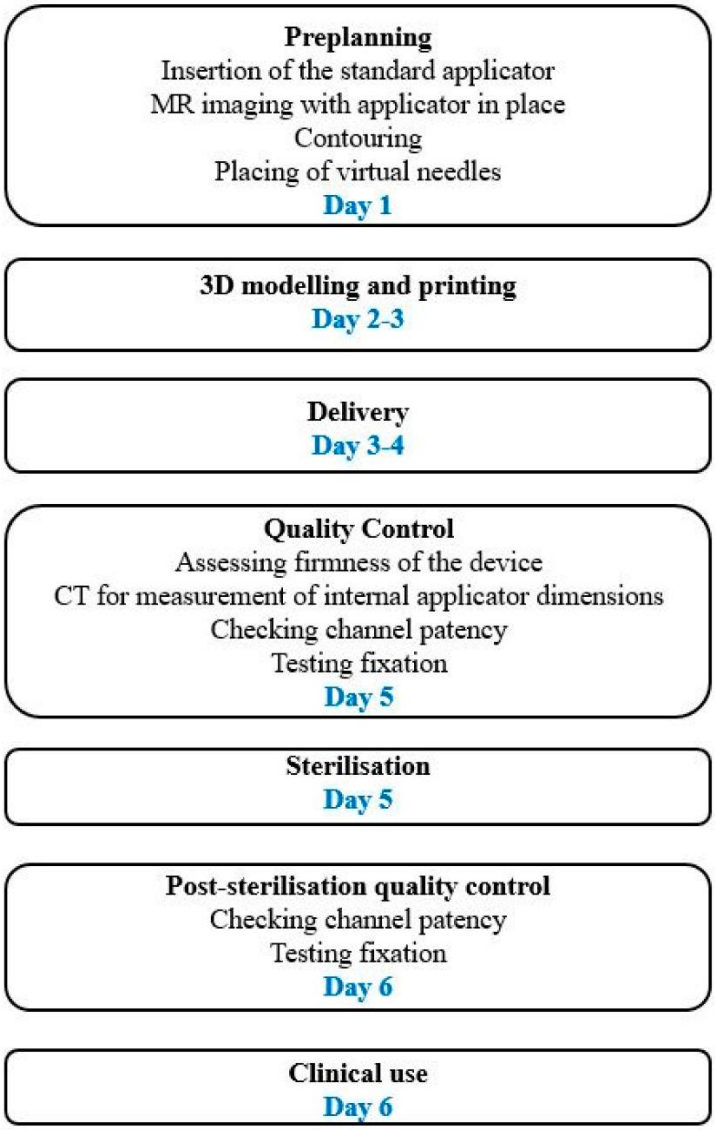


Figure 4: Typical 3-D Brachytherapy process

2.4 Indications for BT: Postoperative Radiotherapy

Surgery, comprising hysterectomy and bilateral oophorectomy, stands as the most crucial treatment for the majority of endometrial cancer patients. For more than half of these patients, no further adjuvant treatment is required, resulting in an outstanding 95% recurrence-free survival rate [Literature Review]. The primary indication for brachytherapy is postoperative, aimed at preventing local vaginal recurrence. Less frequent indications include primary treatment for non-surgical candidates and treating recurrent vaginal disease [41].

2.5 Treatment Protocol, Prescription and Fractionations

The precise administration of treatment for patients grappling with gynecological malignancies hinges upon several variables, encompassing disease pathology, patient condition, and the professional judgment of the radiation oncologist, among other factors. Variables such as the

topographical and histological origin, stage, grade, geometric location of the disease in relation to at-risk organs, prior treatments involving chemotherapy regimens, and radiation play pivotal roles in treatment decisions [421].

Treatment protocols, comprising combinations of radiation and non-radiation therapies, are available for managing endometrial and other malignancies. Often, decisions at this level are deliberated in tumor conferences, involving a consortium of physicians.

Once radiation is incorporated as a chosen treatment, the selection of a modality or modalities becomes pivotal. The ultimate decision on how to employ radiation is made by the radiation oncologist. For instance, options may include HDR brachytherapy alone or a combination of whole pelvic External Beam Radiation Therapy (EBRT) with HDR brachytherapy. After finalizing the radiation modalities, determining the absorbed radiation dose becomes crucial. Radiation dose is measured in grays (Gy). A radiation dose prescription involves consideration of both geometric aspects—such as surfaces, depths, or specific points—and radiobiological factors.

Prescribing the dose involves selecting a surface or point due to the non-uniform nature of dose distribution. For example, in EBRT, dosage is often prescribed to a volume and normalized to the isocenter. In gynecological applications, various methods are employed to prescribe radiation dose. This may include selecting a prescription depth, like the surface of a vaginal cylinder or a distance of 5mm from the cylinder's surface, along with specifying an active length of the cylinder. Alternatively, dose prescription to a specific point is feasible, as seen in the tandem and ovoid applicator as detailed in the International Commission on Radiation Units & Measurements (ICRU) report 38. To compare various prescriptions and modalities and to assess cumulative effects, the concept of biologically effective dose (BED) or equivalent dose in 2 Gy fractions (EQD2) is utilized [11, 43].

Regarding HDR in postoperative brachytherapy, a broad spectrum of schedules exists in the literature, showcasing a wide range of doses when converted to 2 Gy fractions (EQD2), whereby Variations in additional depth introduced by differing schedules include a frequently used regimen of 21 Gy in 3 fractions or 7 Gy prescribed at 5 mm from the surface, as employed in Post Operative Therapy in PORTEC-2. This schedule aims to address the disease with considerable uncertainty surrounding the correct α/β ratio. Previously, a value of 10 has been utilized. However, biologically, the characteristics of this cancer are more akin to breast and prostate cancers, which have demonstrated a much lower α/β ratio. The previously assumed values around 1.5 may not be appropriate, suggesting a figure around 4-6 is more likely. Using an α/β of 4.5 for the tumor, the EQD2 for 21 Gy in 3 fractions is calculated to be 37.2 Gy at 5 mm depth and approximately 150% at the surface, resulting in EQD2 of 55.8 Gy.

Compared to external beam radiotherapy, brachytherapy entails a steeper dose gradient not accounted for in the EQD2 calculation. Considering three Organs at Risk (OARs), the dose at 5 mm is 42.0 Gy and 63 Gy at the surface, explaining the increased rate of mild to vaginal mucosal atrophy compared to external beam radiotherapy in PORTEC-2, equivalent to 4 fractions of 6.0 Gy and 5 fractions of 5.0 Gy. Alternative schedules at 5 degrees in the literature encompass total doses between 15 and 24 Gy administered in 3 to 4 fractions: 3 - 4 x 5—5.5 Gy, 3 - 4 x 6 Gy, corresponding to EQD2 of 21.9 — 38.8 Gy for an α/β of 4.5 and 24 — 43 Gy with an α/β of 3, considering 150% at the surface. The range of EQD2 at the vaginal surface spans between 32.8 and 58.2 Gy for an α/β of 4.5.

The time intervals between fractions vary in the literature. In PORTEC-2, an interval was maintained between each fraction. Due to these variations, different schedules are not directly comparable, even if the EQD2 values are calculated [11, 43].

2.6 Introduction to Tissue Toxicities and Radiation Biology

The fundamental goal of a course of radiation therapy treatment is to deliver the prescribed dose to the tumor while minimizing the impact on normal tissues, thus achieving optimal control and improving survival rates while mitigating undesired side effects. Various organs and tissues, including healthy and tumor tissues, respond differently to radiation, some being more sensitive to side effects than others. Unwanted side effects generally fall into two categories: acute and delayed effects, further categorized as stochastic or threshold effects. Specific scales are utilized to assess unwanted side effects. For instance, the Therapy Oncology Group (RTOG), the Effects of Normal Tissues—Subjective, Objective Management, Analytic (LENT-SOMA), and Adverse Events (CTCAE) v4.0 offer criteria for ranking and classifying rectal responses following radiation treatments [49]. To balance unwanted side effects with tumor control, treatments are fractionated. Evaluation involves both physician experience and quantitative assessment using various dose constraints and plan metrics [43].

Treatment-related toxicities are recorded using the Therapy Oncology Group acute and late morbidity scoring criteria. Acute toxicities for the rectum and sigmoid regions fall under the lower gastrointestinal tract and pelvis category, while late toxicities are assessed based on the small/large intestine. If differentiation between rectal and sigmoid toxicity is challenging, the event is classified as rectal toxicity, considering its higher likelihood to affect the rectum. Events occurring within 90 days of treatment initiation are classified as acute, while others are deemed late toxicities. The most severe score is used for analysis. Patients are categorized into four groups based on toxicity severity: group 1 (grade 0), group 2 (grades 1-4), group 3 (grades 1-3), and group 4 (grades 2-4) in alignment with previous reports [42].

Despite limitations in brachytherapy planning, toxicity can be reliably correlated to dose based on International Commission on Radiological Units and Measurements (ICRU) points. The advent of 3D-based planning has prompted investigations into volume-based dose parameters, which also demonstrate clinically meaningful correlations with toxicities [45].

2.7 Biological Effective Dose (BED) and Equivalent Dose 2 Gy per Fractions (EQD2)

The concept of biologically effective dose (BED) was developed in the early 1980s and has since become commonly utilized in isoeffective dose calculations. Subsequent refinements to the model have expanded its scope, resulting in increased reporting of BED in the publication of clinical results. This facilitates comparisons between cancer centers employing different dose fractionation schedules. For fractionated treatments like External Beam Radiotherapy (EBRT) and High Dose Rate (HDR), with 'n' fractions delivering 'd' dose per fraction and a total dose of 'nd', the biologically effective dose is calculated as:

$$BED = (1 + d/\alpha/\beta)$$

where the parameter α/β is known as the alpha-beta ratio, originating from the established quadratic model in radiobiology as described in reference 5. The alpha-beta ratio depends on the biological effect and the tissue type under consideration, encompassing aspects such as tumor control, acute tissue toxicity, and late normal tissue toxicity effects. Typically, α/β = 10 for acute normal tissue toxicities and α/β = 3 for late normal tissue toxicity effects. When fraction sizes vary, the formula may account for different fraction sizes by summing the individual fractions, such that for 'n' fractions with 'di' doses for the 'i-th' fraction, the BED is given by:

$$BED = \Sigma\, di(1 + d/\alpha/\beta)\ (2).$$

The quantity EQD2 serves as a means to standardize BED concerning a treatment fractionated with a dose of 2 Gy per fraction, a conventional fractionation method at a given time [42, 46]. To achieve this standardization, the BED is simply divided by.

$$EQD2 = BED + 2\alpha/\beta.$$

2.8 Vaginal Cuff Brachytherapy

One of the primary curative treatments for patients suffering from cancer in the endometrium of the uterine corpus is a curative hysterectomy. However, this therapy alone showed an approximate 12% recurrence rate in a study conducted in 1971. The most common site of recurrence is the vaginal cuff, referring to the area where the vagina used to be attached to the uterus. Therefore, the fundamental rationale behind cuff irradiation is prophylactic, aiming to

prevent the recurrence of the disease. Most patients included in this retrospective dosimetric study had Vaginal Cuff Brachytherapy (VCBT) as part of their treatment plan.

2.9 Protocols

In 1978, the International Commission on Radiation Units and Measurements (ICRU) published Report 29 titled 'Dose Specification for Reporting External Beam Therapy with Photons and Electrons' (ICRU, 1978). This current report addresses dose and volume specifications for reporting intracavitary therapy, primarily focusing on cervix carcinoma treatment. However, the concepts presented here are meant to apply to various other intracavitary applications, such as uterine body treatments, etc. Intracavitary therapy is often combined with external beam therapy, hence advocating the use of consistent terminology and definitions.

Due to the high dose-gradient around the sources within the tumor and target volume, specifying the target absorbed dose in terms of the dose absorbed at one or several reference points within the target volume is considered meaningful. Thus, the approach used in external beam therapy cannot be directly applied in this context. The treatment of gynecological malignancies, particularly cervical malignancies, has been one of the earliest applications of radionuclides following Marie Curie's discovery and isolation of radium (226Ra). The use of radium for treating cervical malignancies rapidly became widespread. Over the decades, remarkable advancements in the diagnosis and treatment of such tumors have occurred.

Subsequent to the discovery of radium, intermediate-dose rate, high-dose rate, and pulsed-dose rate radioactive stepping sources were developed, incorporating computer-controlled positioning through afterloading techniques. Currently, three-dimensional treatment planning derived from CT and MR imaging stands as the standard for cervix cancer brachytherapy, replacing the earlier radiographic approach. These techniques are often integrated with advanced external-beam therapy and concurrent cytotoxic chemotherapy using three-dimensional image-guided methods [47].

Numerous advancements have arisen since the publication of ICRU Report 38 (1985) on intracavitary therapy in gynecology. This report elaborates specific concepts and terminology akin to the GEC ESTRO (Groupe Européen de Curiethérapie and the European Society for Radiotherapy & Oncology) contouring of various tumor-related (adaptive) volumes, along with the delineation of adjacent organs at risk (OAR). Detailed delineation of these structures based on repetitive imaging and gynecologic examinations necessitates a comprehensive understanding of tumor spread and response.

This report furnishes comprehensive recommendations for prescribing, recording, and reporting cervix cancer brachytherapy volumetric parameters. However, it is acknowledged that a majority



of advanced cervix cancer patients are, and will be, treated in developing countries. Hence, the concepts and terms described in this report follow the tradition of previous ICRU reports on prescribing, recording, and reporting external beam radiotherapy (from ICRU Report 50 to 83). Specifically, the report elaborates on concepts for Gross Tumor Volume (GTV), Clinical Target Volume (CTV), and Planning Target Volume (PTV) within the context of cervix cancer, highlighting a new adaptive target concept.

To address the needs of image-guided brachytherapy, dose-volume parameters are introduced, incorporating changes in tumor topography into these concepts and terms. However, challenges remain in assessing volumes and doses from External Beam Radiotherapy (EBRT) and integrating them into the overall treatment timeline. The second part of the GYN GEC ESTRO (Groupe Européen de Curiethérapie and the European Society for Radiotherapy & Oncology) recommendations focuses on 3D dose-volume parameters for brachytherapy of cervical carcinoma [48].

Methods and parameters have been developed and validated based on dosimetry, imaging, and clinical experience from various institutions (University of Vienna, Gustave Roussy Paris, University of Leuven). Cumulative dose-volume histograms (DVH) are recommended for evaluating complex dose heterogeneity. DVH parameters for GTV, HR CTV, and IR CTV include the minimum dose delivered to 90% and 100% of the respective volume (D90, D100). Additionally, volumes enclosed by 150% or 200% of the prescribed dose (V150, V200) are recommended for assessing high dose volumes, with V100 serving for quality assessment only within a given treatment schedule.

Regarding Organs at Risk (OAR), the minimum dose in the most radiated tissue volume is recommended for reporting: 0.1, 1, and 2 cm3, optionally 5 and 10 cm3, based on certain assumptions. These assumptions encompass the full dose of external beam therapy in the volume of interest, identical location during fractionated contiguous volumes, and contouring of organ walls for 0.2 cm3. Dose values are reported as absorbed dose, considering different dose rates.

The linear-quadratic radiobiological model—equivalent dose (EQD2)—is applied for brachytherapy and dose calculation from external beam therapy. This formalism allows for systematic assessment within a patient, a single center, and comparison between different centers, analyzing dose-volume relations for GTV, CTV, and OAR. Recommendations facilitating the transition from traditional to 3D image-based cervix cancer brachytherapy are presented [49].



Chapter-3 Materials & Method



3.1 Materials

3.1.1 HDR brachytherapy treatment unit

The use of HDR brachytherapy as an integral part of cancer treatment spans nearly a century. Its advancements have been propelled by the development of afterloading devices and novel radioisotopes. Remote afterloading significantly enhances radiation control and offers technical benefits, including optimization of isodose distribution, thereby improving patient care. Presently, brachytherapy (BT) is marked by technical innovations, including:

(a) Increasing the number of channels in remote afterloading units.

(b) Development of radioactive microsources.

(c) Creation of new algorithms for computerized treatment planning and dosimetry.

(d) Advancements in imaging techniques and introduction of sectional imaging.

This study utilized the Varian GammaMed remote HDR brachytherapy afterloader, a three-channel system expandable to 24 channels, providing a user-friendly approach to initiating HDR treatments. It seamlessly integrates with the BrachyVision™ treatment planning system and the ARIA oncology information system, facilitating connectivity for up to 24 channels.

- A solid-core source cable enabling movement from distal to proximal directions, enhancing accuracy and reducing errors from cable bunching.
- Standardized length of the applicator and source guide tube at 1,300 mm, improving patient safety and reducing errors associated with incorrect treatment cable length.
- The distal 200 mm of the cable boasts a highly flexible braided design, enabling smooth navigation through tight catheter turns.
- The device offers 60 programmable dwell positions per channel, with step sizes ranging from 1 mm to 10 mm.

Adherence to international standards of safety and quality, such as those outlined by the International Electrotechnical Commission or the International Organization for Standardization (ISO 9000 131), is imperative for HDR remote afterloading systems.

3.1.2 High dose rate radioactive source

To achieve the necessary High-Dose Rate (HDR) and source size required for both intracavitary and interstitial Brachytherapy (BT), a radioisotope with high specific activity is essential. Iridium-192 stands out in BT applications due to its high specific activity (330 MBq/g), relatively low gamma energy (averaging 0.375 MeV), and a comparatively short half-life (73.8 days). Presently, most High Dose Rate (HDR) remote afterloaders employ a single source of



approximately 370 GBq. This source features an active length of about 3.5 mm and an active diameter of 0.5 mm. Encased within, the source measures approximately 5 mm (with some sources reaching up to 10 mm) in length and less than 1.5 mm in diameter. These dimensions may vary across different commercial models.

The source is intricately connected to the end of a drive cable, allowing its transfer to programmed positions within the applicators (known as dwell positions) and its secure placement for the specified duration (dwell times) through a motor-driven system.

3.1.3 Afterloader device

Remote afterloading brachytherapy systems offer an automated means to deliver a radioisotope directly to cancerous tissue, thereby reducing radiation exposure to surrounding healthy tissue and eliminating the risk of radiation exposure to hospital staff [Smith, et al., Literature Review]. The dosage of radiation administered varies depending on the chosen brachytherapy method. Low-dose-rate (LDR) brachytherapy involves an implanted source that delivers a dose of 40 to 60 centigrays (cGy) per hour over several days. In contrast, high dose-rate (HDR) brachytherapy employs a traveling (stepping) source that delivers a dosage higher than cGy per minute for 5 to 30 minutes.

Once treatment parameters are established, a source drive mechanism, typically a computer-controlled stepper motor with drive rollers or belts, propels the source from the shielded safe through guide tubes into the treatment applicators. These guide tubes, also known as transfer tubes, ensure precise placement of the source within the applicators. Additionally, the indexer, usually providing 18 to 24 channels, facilitates the entry and transfer of the source for complex treatments requiring multiple applicators.

An afterloader device, being mobile and space-efficient, necessitates specific design elements, including:

(a) A shielded safe (main source container) to house the source when not in use;

(b) A stepping motor;

(c) A system for feedback on source transfer and positioning;

(d) Multiple channels for source transport;

(e) An indexer for automatic transfer of the source cable between different transfer tubes connecting the device to the applicators.

3.1.4 Control Console

The control console, positioned outside the treatment room, manages the afterloader operations and displays treatment data. The treatment plan can be transmitted to the control console from the planning computer via DICOM. Equipped with a microprocessor, the control console automatically adjusts dwell times to account for radioactive decay, simplifying operational procedures.

3.1.5 Applicators

Almost all applicators initially designed for Low-Dose Rate (LDR) manual afterloading have undergone redesigns for High-Dose Rate (HDR) use, integrating mechanisms to link them to the afterloader device via transfer tubes. These connections incorporate mechanical interlocks to ensure precise positioning and attachment of the applicator to the transfer tubes, effectively preventing any incorrect connections. Notably, HDR applicators commonly feature thinner tubes. Once interconnected, the applicator, transfer tubes, and afterloader device create a closed system, preventing any potential dislodgment of the radiation source within the patient's body or its unintended exposure outside the target region before reaching its intended location.

These HDR applicators come in three main categories: intracavitary, intraluminal, and interstitial, each with a distinct connector or transfer tube tailored for connection to the treatment unit. The specific design of transfer tubes for intracavitary applicators ensures uniform overall length while incorporating different interlocks for each treatment channel, minimizing connection errors. The diversity of intracavitary applicators available for HDR treatment spans various materials and construction. Some applicators, crafted from stainless steel, prioritize durability and compatibility with X-ray simulation. Conversely, others, constructed from materials like carbon fiber, avoid producing artifacts on Computed Tomography (CT) or Magnetic Resonance Imaging (MRI) scans.

Certain intracavitary applicators, such as the Fletcher type, possess a rigid, unfixed geometry, necessitating tailored treatment planning for individual patients. Conversely, fixed geometry applicators like ring applicators allow for standardized dose distribution planning before insertion, while still permitting dwell time optimization to accommodate 3-D image-guided personalized dosimetry.

3.1.5.1 Single Channel Applicator

Most treatments involve intracavitary vaginal brachytherapy using a single-channel cylinder due to its simplicity and resource efficiency compared to colpostats or vaginal molds.

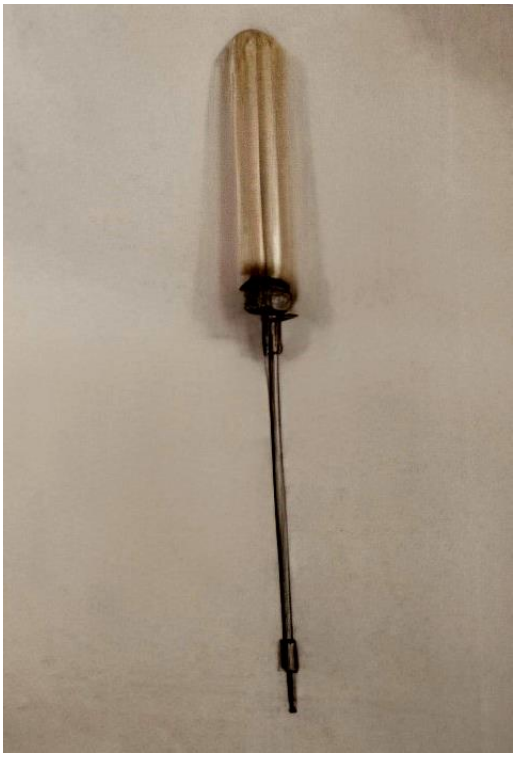


Figure 5: Single Channel Applicator.

However, this applicator's radial dose distribution limits opportunities for optimizing the treatment plan, potentially leading to air gaps on the vaginal surface. This can increase hotspots, especially smaller than the apex, which could heighten toxicity levels.

The use of a multi-channel cylinder presents advantages by enabling better conformal dosimetry and reducing doses to the bladder and rectum. Efforts to enhance vaginal brachytherapy capabilities have led to the development of multi-channel applicators.

3.1.5.2 Miami Multichannel Applicator

Multichannel applicators (MCs) offer the advantage of selectively treating localized disease while minimizing radiation exposure to Organs at Risk (OARs) by adjusting positions along the channels. Compared to single-channel applicators (SCs), MCs exhibit steeper radial dose gradients as the source channels approach the tumor volume. Moreover, the strategic loading of channels aims to improve dose homogeneity, minimizing OAR exposure while delivering the intended dose to the lesion being treated.

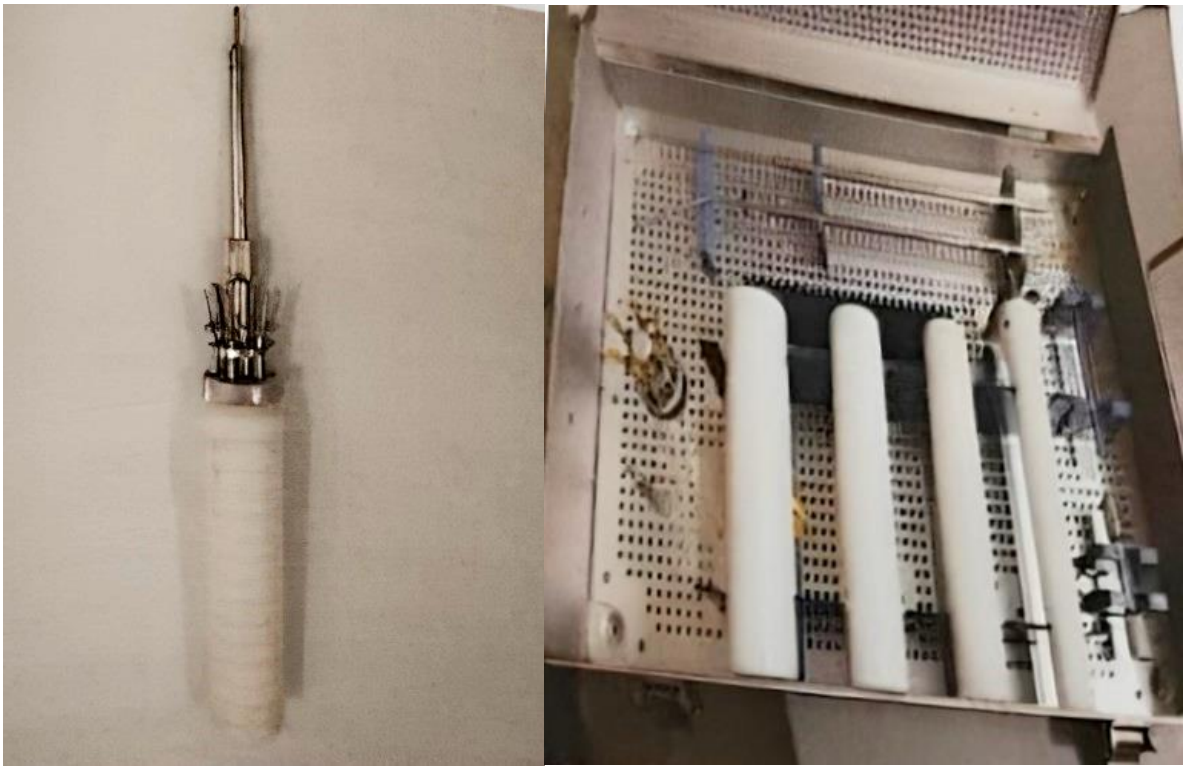


Figure 6: Miami Multichannel Applicator

In the pursuit of enhancing vaginal brachytherapy capabilities, multichannel applicators have been developed. These additional channels, situated at the periphery of the applicator, facilitate more precise dosimetry, mitigating the anisotropy arising from a single line source at the vaginal apex. The differential nature of these channels potentially reduces bladder and rectal doses in comparison to the single-channel cylinder.

3.1.6 Treatment Planning System (TPS)

The Treatment Planning System (TPS), an integral component of a treatment unit, facilitates dose calculations swiftly, transferring the data to the BrachyVision treatment planning system for plan development. This system offers a comprehensive toolset ensuring consistency and efficiency in the planning process. BrachyVision enables customization of plans according to individual patient needs and disease characteristics. The system includes:

(a) Input devices for simulation images (e.g., digitizer), CT/MRI scans;

(b) Control for source strength input;

(c) Reconstruction of source channel;

(d) Algorithm for placement of source dwell positions;

(e) Graphic visualization of implantation and 3D visualization;

(f) Dwell time calculations;

(g) Dose distribution algorithm;

(h) Option for dose volume histograms calculation;

(i) Potential for optimization;

(j) Calculation representation and display method (e.g., printer);

(k) Method for transferring the plan to the treatment unit.

3.2 Methods

3.2.1 patient Evaluation

Women diagnosed with locally advanced Gynecology and Obstetrics (FIGO) stage IB2-IVA cervical cancer undergo definitive treatment after a biopsy for histopathological diagnosis. The FIGO work-up protocol mandates a comprehensive assessment under anesthesia, including a pelvic examination, sigmoidoscopy, and cystoscopy. Essential laboratory investigations encompass a complete blood count, basic metabolic panel, liver function tests, and an evaluation for the presence of hydroureter or hydronephrosis indicating possible direct tumor invasion or ureteral obstruction due to metastasis. Intervention such as stent placement or nephrostomy tube insertion might be necessary to alleviate significant urinary obstruction prior to initiating therapy.

Before the commencement of vaginal cuff brachytherapy (VCBT), the patient undergoes a consultation with a radiation oncologist. During this consultation, the oncologist discusses the advantages and potential risks associated with the treatment and addresses any queries or concerns the patient may have. Once the patient consents to receive brachytherapy, the treatment sessions are scheduled accordingly.

3.2.2 Patient Selection

This study utilized a dataset comprising 20 consecutive patients undergoing intracavitary brachytherapy for cervical carcinoma at the National Institute of Cancer Research Hospital in Dhaka, Bangladesh. All these patients were classified under the International Federation of Gynecology and Obstetrics (FIGO) stage II. The determination of target and organ at risk volumes followed established systematic guidelines [1, 2].

Initially, all patients received external beam radiotherapy, delivering a dose of 50 Gy in 25 fractions. Subsequently, following a 7-10 day interval, intracavitary brachytherapy was administered in two equally divided fractions, totaling a dose of 14 Gy.

3.2.3 Applicator Specifications

The Segmented Cylinder Applicator Set serves the treatment of vaginal and stump cancer, along with rectal cancer. Depending on the selected front segment, the source dwell position can either be at the front's center for wall treatment or at the distal position for vaginal stump treatment. This segmented structure of the applicator enables easy adaptation to each patient's anatomy, ensuring compatibility with CT scans for the vagina, vaginal stump, and rectum. The set includes conditional 1.5 and 3 Tesla segments, allowing centered first source position for front segments and distal first source position for shaping segments. Various diameters of shaping segments facilitate seamless adaptation to patient anatomy, ensuring homogeneous dose delivery parallel to the cylinder wall.

The Miami Multichannel Applicator set, designed to treat vaginal or cuff diseases, optimizes the dose according to individual patient anatomy and pathology. It comprises a 22-cylinder obturator with six channels encircling it, alongside a single 30° central uterine tandem. Additionally, obturator sleeves in diameters of 30, 35, and 40 mm and four buildup caps (25, 30, 35, and 40 mm in diameter) are included. This applicator can be used with or without a tandem and enhances traditional vaginal or cervical applicators by generating asymmetrical dose distributions along the cylinder's long axis, conforming more accurately to the target area. The set is supplied with an aluminum sterilization tray, and digital radiographic imaging is facilitated by dummy wires inserted into the peripheral channels.

Key features include suitability for vaginal, vaginal cuff, and cervix treatments, compatibility with CT scans, easy transfer guide tube connection through the ClickFit™ connection, asymmetrical dose distribution capabilities, multiple obturator sizes, straight and angled tandems for intrauterine and/or vaginal treatments, and steam sterilizability with an included aluminum sterilization tray. Applicator insertion techniques are essential for anatomically correct placement and extent determination, often performed under general or spinal anesthesia, with considerations for postoperative pain management. Adjustments during imaging aid in pain control during the applicator's dwell duration.

3.2.4 Applicator insertion techniques

For optimal treatment, platinum or carbon tubes are inserted into residual gross disease areas or used to delineate and reduce CT simulation artifacts caused by steel needles. They aid in

target volume-based planning and are recommended for vaginal lesions, accompanied by a perineal template featuring a vaginal cylinder and an array of needles.

3.1.5 Simulation

Prior to the initial treatment insertion, computed tomography (CT) imaging using a Phillips Healthcare system from Amsterdam, the Netherlands, was conducted with the applicator in place. The CT scans, with a slice thickness of a few millimeters, aimed to assess the viability of the technique. Tumor response was evaluated and compared against External Beam Radiation Therapy (EBRT), and a delineation of the high-risk clinical target volume (HRCTV) was conducted using the GECESTRO formalism. Subsequent CT imaging was carried out before each treatment insertion. To facilitate imaging, 60 mL of fluid was introduced into the patient's bladder. Each CT scan was meticulously reviewed to ensure the balloon adhered properly to the vaginal mucosa, eliminating any air gaps. Treatment planning was then executed based on the acquired CT dataset for each insertion.

3.1.6 Contouring Guidelines

The treatment planning for Intracavitary Brachytherapy (ICBT) involved delineating volumes of interest using a 64-slice CT for each session. At the time of each session, the oncologist delineated the clinical target volume (CTV) according to GEC ESTRO guidelines. This CTV encompassed the high-risk areas of the uterus, extending beyond the recommended structure size by GEC ESTRO.

Once the volumes of interest were delineated and the dose algorithm was executed to calculate optimal dwell times and doses, certain constraints were set for different anatomical structures:

- For CTV: 70% of the prescribed dose was set to be received by 100% of the volume, with 7 Gy delivered to 100% of the volume.
- For the rectum: Constraints were set at 80% of the prescribed dose, with 5.6 Gy to 5% of the volume and 4 Gy to 100% of the volume.
- For the bladder: Limits were set at 70% of the prescribed dose, with 4.9 Gy to 5% of the volume and 3.5 Gy to 100% of the volume.

These constraints for organs at risk were prioritized at 50% relative to the CTV. Additionally, the dwell time constraints were fixed at 3 milliseconds.

A total dose of 14 Gy in two daily fractions of 7 Gy delivered over one week was prescribed for the CTV. Due to the spread of CT images, an evaluation was necessary to assess the impact of 3D imaging on dosimetry. Variations in doses were observed based on the delineation

method. Typically, comparisons of International Commission on Radiation Units and Measurements (ICRU) D2cc doses on 3D images were prescribed to the applicator surface or at a depth of 0.5 cm. This choice was based on the vaginal wall thickness and lymphatic vessel placement.

Approximately 50% of lymphatic channels were found 1 mm beneath the mucosa and 7% within 3-4 mm. A vast majority (95%) of vaginal lymphatic channels resided within the first 3 mm from the vaginal surface. Hence, the necessity to personalize the prescription depth according to individual variations in vaginal thickness was highlighted.

3.2.7 Dose Prescription

A total dose of 14 Gy in two daily fractions of 7 Gy delivered over 1 week was prescribed for the Clinical Target Volume (CTV). However, the variation seen in CT images prompted an evaluation of the impact of 3D image dosimetry. Studies have highlighted dose variations depending on catheter placement methods. Typically, comparisons of International Commission on Radiation Units and Measurements (ICRU) of D2cc doses on 3D images were usually referenced to the applicator surface or at a depth of 0.5 cm. This choice considers the thickness of the vaginal wall and the depth placement of lymphatic vessels.

Approximately 50% of lymphatic channels are located within 1 mm beneath the mucosa, with 7% situated within 3-4 mm. Notably, 95% of vaginal lymphatic channels are positioned within the first 3 mm from the vaginal surface. Individualizing the prescription depth based on vaginal thickness has been explored to minimize toxicity. Comparisons between surface, mucosa, and various depths in comparison with the 0.5 cm depth dose target revealed dose variations up to 172% for the 0.5 cm depth prescription. This underscores the need for precise and patient-specific High-Dose Rate (HDR) Vaginal Cylinder Brachytherapy (VCBT) planning to ensure organs at risk receive optimal doses and avoid overdoses.

3.2.8 Treatment planning

A customized patient specific treatment plan is made for each fraction of each treatment in the course of an HDR VCBT plan for each patient. VCBT treatment planning involves several key stages: contouring, catheter reconstruction, dose prescription, and optimization. Following the export of CT simulation images into the Brachyvision TPS (Treatment Planning System), contouring of organs at risk is conducted by oncologists. Digitally reconstructed images guide the graphical construction of catheters through which the Ir-192 source travels within the vaginal cylinder applicator. Once catheter contouring is completed, dose prescription is implemented and calculated using the TG-43 dose calculation formalism. Notably, the TG-43

formalism models the human body akin to water medium and does not factor in tissue heterogeneities.

To attain an optimally tailored dose distribution, an Inverse Planning Simulated Annealing (IPSA) algorithm within the Brachyvision TPS is employed. This algorithm utilizes dose volume constraints to calculate dwell positions and times of the Ir-192 source within the catheters, thereby achieving an ideal patient-specific dose distribution. After optimization, a secondary calculation validates selected points to ensure accurate dose computation by the TPS. Subsequently, a medical physicist reviews the plan before the radiation oncologist conducts a final verification, sometimes making minor adjustments. Upon approval, the plan is exported for implementation using the treatment delivery system, thereby executing the High Dose Rate treatment.

3.2.9 Treatment Delivery

Once the treatment plan is imported into the computer system controlling the remote afterloader, the radiation therapist ensures its accuracy. Subsequently, the therapist connects tubes from the remote afterloader to those of the patient. A preparatory 'dry run' involving an inactive source is performed to confirm that the source moves smoothly within the catheters and applicator while not emitting radiation to the patient.

Upon completion of the dry run, the door to the shielded room is secured, and the actual treatment is remotely administered by the radiation physicist or therapist who remain outside the treatment room. A typical treatment session lasts between 5 to 15 minutes. Following the treatment, the connections between the afterloader and the patient's tubes are disconnected, and the applicator is removed from the patient. Subsequently, the patient can return home.

3.2.10 Treatment Plan Evaluation

3.2.10.1 Dose-Volumetric Analysis

The dosimetric evaluation involves various parameters: the minimum dose given to 95% of the target volume (D95), the minimum dose to the entire target volume (D100), and the volume covered by the prescribed dose (V100). For the OARs, parameters such as the dose to the most exposed 2 cm^3 volume (D2cc), 0.1 cc, D1cc, V70, and V80 were assessed. Statistical analysis was conducted using a paired sample test in SPSS with a 95% confidence interval.

Both qualitative and quantitative comparisons were made using dose-volume histograms (DVHs) for different dose volumes and DVH parameters. Various volumetric quantities such as dose homogeneity index (DHI), overdose volume index (OI), dose non-uniformity ratio (DNR), and conformity index (COIN) were estimated based on the cumulative DVH. These

parameters aid in evaluating aspects like overdose, uniformity, and conformity of the prescribed dose to the target volume while considering non-target structures. Ideal values for these indices (DHI, OI, DNR, COIN) serve as benchmarks for assessing treatment plan quality and conformity to the intended target volume [6-8].

Dose Homogeneity Index (DHI): It represents the fraction of the target volume that receives a dose within the range of 1.0 to 1.5 times the prescribed dose (V100-150).

Dose Overdose Volume Index (OI): This index signifies the fraction of the target volume that receives a dose equal to or greater than 2.0 times the reference dose. It's calculated as the ratio of the volume receiving 200% of the prescribed dose to the volume receiving 100% of the prescribed dose (OI = V200/V100).

Dose Non-Uniformity Ratio (DNR): It measures the portion of the target volume that receives a dose equal to or greater than 1.5 times the reference dose compared to the volume receiving the reference dose (Ideal DNR = 0). DNR is expressed as the ratio of V150 to V100 (DNR = V150/V100).

Conformity Index (COIN): COIN reflects how well the prescribed dose covers the target volume while avoiding non-target structures (Ideal COIN = 1). It's determined by V100 not covering the volume of non-target critical structures (CTV).

3.2. BED and EQD2 (Biologically Equivalent Dose and Equivalent Dose)

BED is the sum of biologically equivalent doses from External Beam Radiation Therapy (EBRT) and HDR Brachytherapy. EQD2 can be computed using the formula (2), where 'n' signifies the number of fractions, 'd' represents the dose per fraction, and an alpha/beta (α/β) ratio of 10 Gy for tumors and 3 Gy for critical structures is considered. EQD2 values differ when the α/β ratio is 10 (EQD2 Gy10) and 3 Gy (EQD2 Gy3) [27].

$$\text{Cumulative BED} = \text{BED(EBRT)} + \text{BED}_{\text{HDR}}$$

$$\text{BED} + \sum n \cdot d \cdot \left(1 + \frac{\alpha}{\beta d}\right)$$

$$\text{EQD2} = \left\{ d \cdot \left(1 + \frac{\alpha}{\beta d}\right) \right\} / 2$$

3.7 Source Characterization

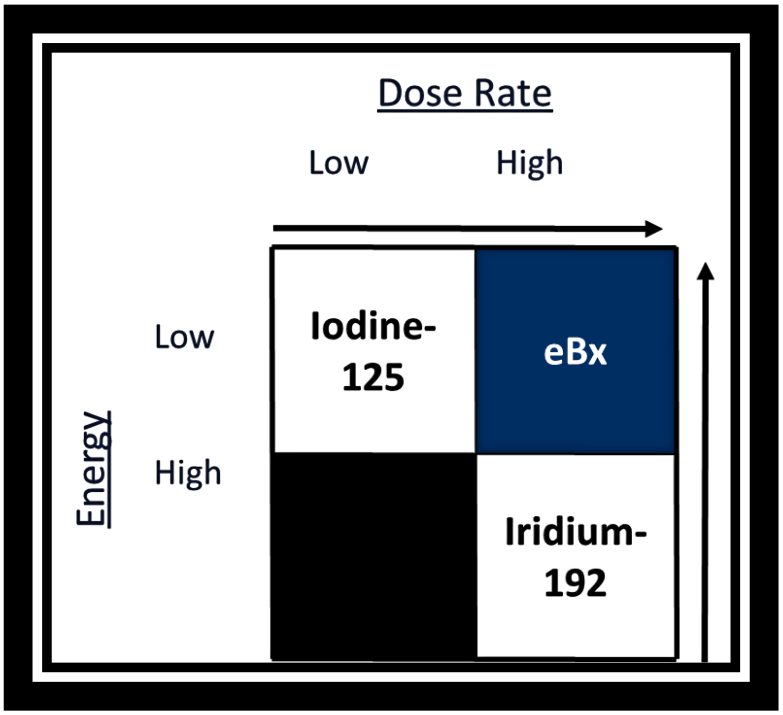


Figure 7: Source Characterization

Optimal combination

not found in any current

medical isotope

High dose rate

Outpatient treatment times same as fresh Ir-192 source

Low energy

Energy adequate to penetrate target tissue. Minimal stray radiation to normal tissue, organs, and medical personnel Requires less shielding.

3.8 Prescribed dose

Prescribed Dose: 700 cGy

Treatment: Vaginal



Chapter-4 Results & Discussion

4.1 Results & Discussions

4.1.1 Dose Analysis of Bladder and Rectum for single channel

The dose distribution of single channel is showed in Fig 23 and dose distribution of three channel electronic brachy therapy is showed in Fig 24 below.

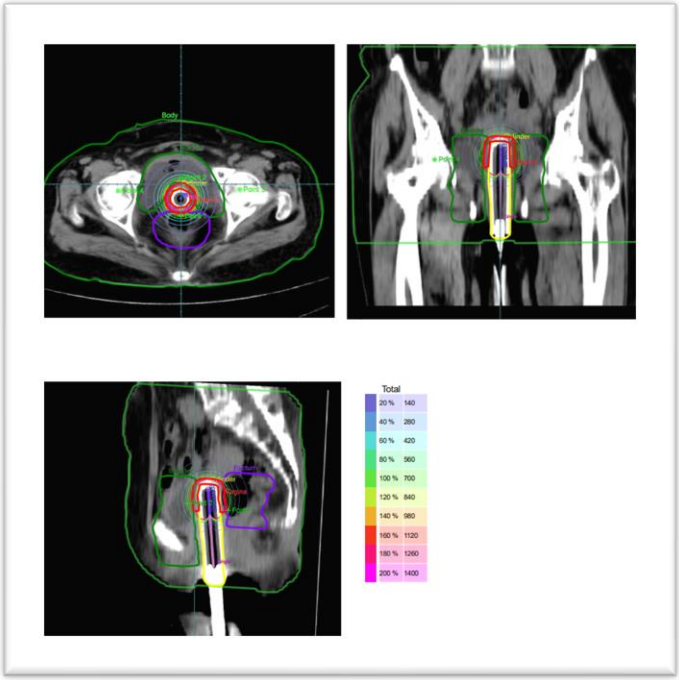


Figure 8: Single channel

The comparison of organ doses, particularly the vagina, bladder, and rectum, within the context of prescribed constraints, yields critical insights into the efficacy and adherence of HDR electronic brachytherapy.

Vagina Analysis:

DV% Constraint vs. Value: The V90% aimed for >95% was effectively achieved at 100.0%. Similarly, the D98% and D90% constraints were set at 939 cGy and 1131 cGy, respectively.

Dose Distribution (Dmax, D.Med, D.Min): The vagina exhibited a considerable discrepancy in doses. The maximum dose (Dmax) recorded was 4387.4 cGy, markedly surpassing the intended 700 cGy. Median (D.Med) and minimum (D.Min) doses were 1816.2 cGy and 782.1 cGy, respectively. These values indicate a substantial dose escalation within the vaginal region.

Bladder Analysis:

DV% Constraint vs. Value: The D2 cm^3 constraint (<9000 cGy) had an observed value of 1045 cGy, indicating a dose lower than the specified limit. However, the D0.1 cm^3 constraint



(<11000 cGy) recorded a value of 1353 cGy, exceeding the intended limit. Dose Distribution (Dmax, D.Med, D.Min): The bladder received elevated doses compared to the prescribed limit of 700 cGy. The maximum dose (Dmax) reached 1541.0 cGy, significantly surpassing the prescribed dose. Median (D.Med) and minimum (D.Min) doses were 138.9 cGy and 9.2 cGy, respectively.

Rectum Analysis:

DV% Constraint vs. Value: The D2 cm³ constraint (<7500 cGy) obtained a value of 804 cGy, indicating a dose lower than the specified limit. However, the D0.1 cm³ dose was 1411 cGy, which was not explicitly constrained but was notably high.nDose Distribution (Dmax, D.Med, D.Min): The rectum, like the bladder, experienced doses well above the prescribed 700 cGy. The maximum dose (Dmax) recorded was 1853.0 cGy, significantly surpassing the intended dose. Median (D.Med) and minimum (D.Min) doses were 166.1 cGy and 23.8 cGy, respectively. While the bladder and rectum showed variations in adhering to prescribed constraints, both organs received doses substantially exceeding the intended levels. These escalated doses indicate potential risks of adverse effects or complications, emphasizing the need for meticulous treatment planning and dose optimization to ensure safer and more effective brachytherapy procedures for these organs.

Table 4.1 Dose Analysis of Bladder and rectum for Single channel

Organ	DV% Constraint	DV% Value	Organ	Dmax (cGy)	D.Med (cGy)	D.Min (cGy)
Vagina	V90% > 95%	100.0%	Vagina	4387.4	1816.2	782.1
Vagina	D98%	939 cGy	Bladder	1541.0	138.9	9.2
Vagina	D90%	1131 cGy	Rectum	1853.0	166.1	23.8
Bladder	D2 cm³ < 9000 cGy	1045 cGy				
Bladder	D0.1 cm³ < 11000 cGy	1353 cGy				
Rectum	D2 cm³ < 7500 cGy	804 cGy				
Rectum	D0.1 cm³	1411 cGy				

4.1.2 Dose Analysis of Bladder and Rectum for three channel

The comparison of bladder and rectal doses using three-channel vaginal applicators in HDR electronic brachytherapy reveals crucial insights.

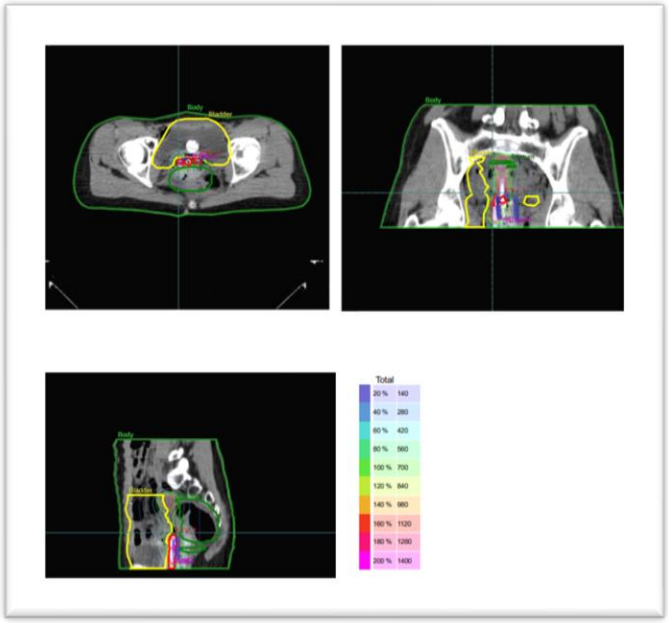


Figure 9: Three channel

Bladder Analysis:

DV% Constraint vs. Value: The prescribed constraint for D2 cm³ (<9000 cGy) was exceeded, with a value of 1148 cGy, indicating a higher dose received than anticipated. Dose Distribution (Dmax, D.Med, D.Min): The maximum dose (Dmax) received by the bladder was 2943.8 cGy, significantly higher than the prescribed 700 cGy. Median (D.Med) and minimum (D.Min) doses were 162.6 cGy and 18.3 cGy, respectively. D1 cm³ and D0.1 cm³: The values for these constraints were 1378 cGy and 2228 cGy, respectively. These doses demonstrate varying degrees of escalation beyond the prescribed limits.

Rectum Analysis:

DV% Constraint vs. Value: Similar to the bladder, the prescribed D2 cm³ constraint (<7000 cGy) was surpassed with a value of 767 cGy. Dose Distribution (Dmax, D.Med, D.Min): The maximum rectal dose (Dmax) recorded was 3193.9 cGy, substantially higher than the prescribed 700 cGy. The median (D.Med) and minimum (D.Min) doses were 171.6 cGy and 34.0 cGy, respectively. D1 cm³ and D0.1 cm³: The doses for these constraints were 897 cGy and 1614 cGy, indicating notable deviations from the intended doses. Both the bladder and rectum experienced dose escalations beyond the prescribed constraints across various measured parameters. The maximum doses recorded for both organs far exceeded the intended dosage of 700 cGy, suggesting a heightened risk of adverse effects or complications.

Table 4.2 Dose Analysis of Bladder and rectum for Three channel

Organ	DV% Constraint	DV% Value	Organ	Dmax (cGy)	D.Med (cGy)	D.Min (cGy)
Bladder	D2 cm³ < 9000 cGy	1148 cGy	Bladder	2943.8	162.6	18.3
Bladder	D1 cm³	1378 cGy	Rectum	3193.9	171.6	34.0
Bladder	D0.1 cm³	2228 cGy				
Rectum	D2 cm³ < 7000 cGy	767 cGy				
Rectum	D1 cm³	897 cGy				
Rectum	D0.1 cm³	1614 cGy				

4.1.3 Dose Volume Histogram

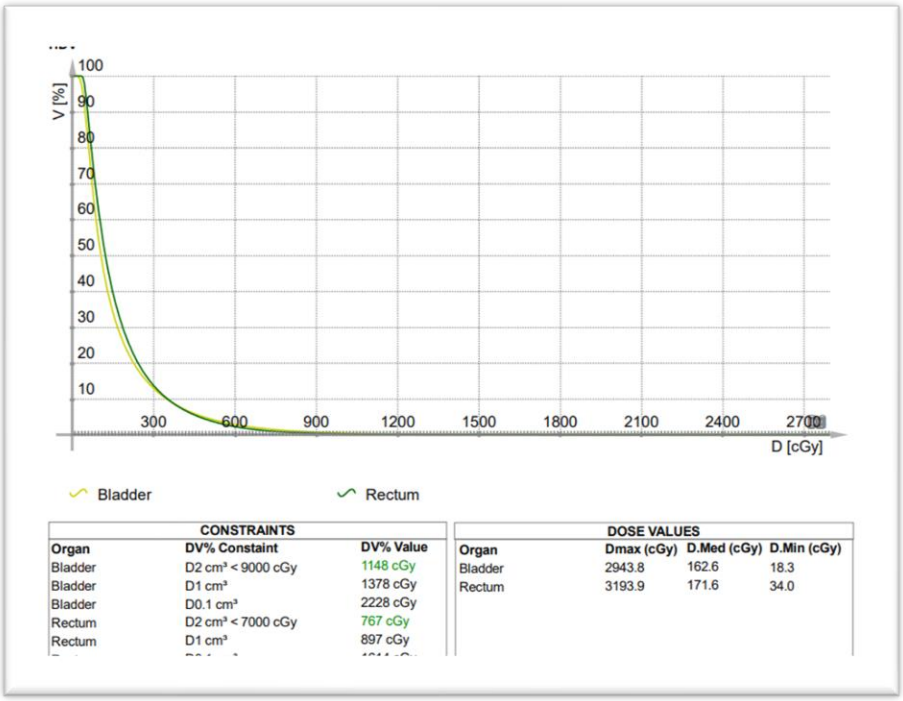


Figure 10: Dose Volume Histogram of three channel

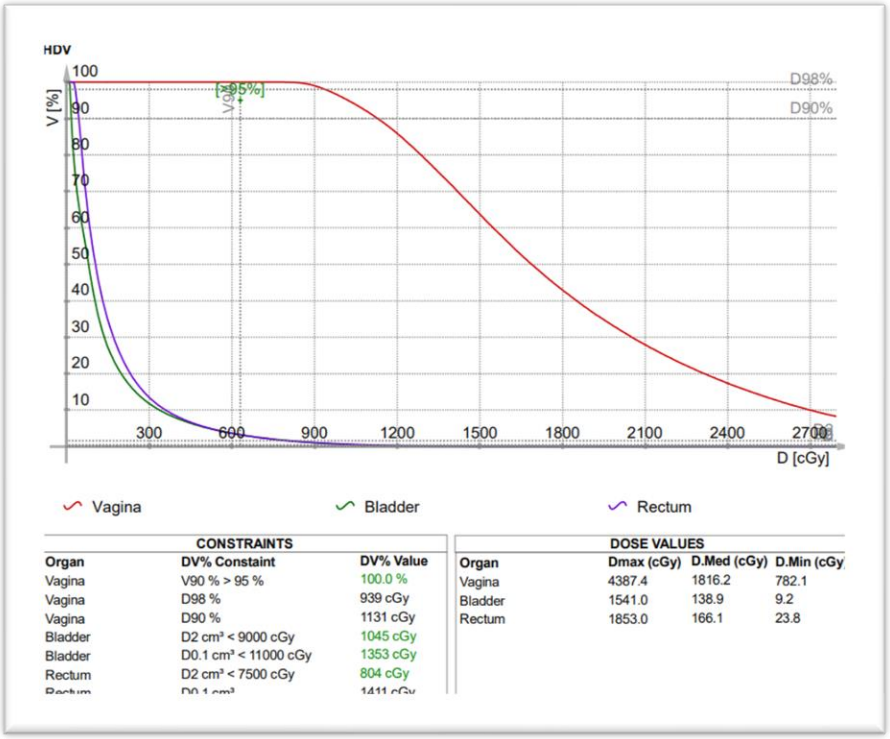


Figure 11: Dose Volume Histogram of single channel



Chapter-5 Conclusion

5.1 Conclusion

The comparison between single-channel and three-channel vaginal applicators in HDR electronic brachytherapy, particularly concerning the bladder and rectum doses, sheds light on crucial aspects of treatment efficacy and adherence to prescribed constraints. The analysis reveals substantial deviations from prescribed constraints across all organs assessed. The vagina, bladder, and rectum received doses far exceeding the intended levels. The vagina experienced considerable dose escalation, with maximum doses surpassing 4000 cGy, well above the prescribed 700 cGy. Both the bladder and rectum showed doses significantly higher than the prescribed limits, highlighting potential risks of adverse effects or complications. The bladder received maximum doses exceeding 1500 cGy, while the rectum registered doses exceeding 1800 cGy, indicating a clear need for meticulous treatment planning and optimization to ensure safer procedures for these organs. With the three-channel applicator, similar patterns of dose escalation were observed in the bladder and rectum. The bladder recorded a maximum dose of nearly 3000 cGy, surpassing the intended 700 cGy by a substantial margin. Likewise, the rectum exhibited maximum doses exceeding 3000 cGy, significantly higher than the prescribed limits. Both organs showed varying degrees of dose escalation beyond the intended constraints, emphasizing the critical need for dose optimization to mitigate potential complications and ensure safer treatment outcomes. The comparison between single and three-channel vaginal applicators revealed consistent trends of escalated doses in the bladder and rectum, indicating a heightened risk of adverse effects. These findings underscore the necessity for comprehensive treatment planning and optimization to align delivered doses more closely with prescribed constraints, ultimately improving the safety and effectiveness of HDR electronic brachytherapy for patients undergoing treatment in these anatomical regions.

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